Environmental Product Declaration

In accordance with ISO 14025:2006, ISO 21930:2017 and EN 15804:2012+A2:2019/AC:2021 for:

Standard Prestressed Steel

from

Fapricela – Indústria de Trefilaria, S.A.

Programme:	The International EPD® System, www.environdec.com
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An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com

General information

Programme information

Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
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Accountabilities for PCR, LCA and independent, third-party verification
Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): *PCR 2019:14 Construction products (EN 15804+A2), version 1.3.3*
PCR review was conducted by: *The Technical Committee of the International EPD® System.*
Life Cycle Assessment (LCA)
LCA accountability: *Rita Pinheiro Garcia, Itecons – Institute for Research and Technological Development in Construction, Energy, Environment and Sustainability*
Third-party verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ EPD verification by individual verifier Third-party verifier: *Elisabet Amat, GREENIZE* Approved by: The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.

Company information

Owner of the EPD:
Fapricela – Indústria de Trefilaria, S.A.

Contact:
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Description of the organisation:
Fapricela – Indústria de Trefilaria, S.A., is a family-owned company founded in 1977. The company began operating as a construction nail manufacturer. However, as the years passed and the company grew, new products were progressively added to the portfolio as a result of the changing market demands, such as smooth steel wires (cold drawn, galvanized, black annealed, galvanized PVC coated and barbed wire), chain link fencing (galvanized, PVC coated and farm fencing), electrowelded mesh. The greatest and most important impulse was during an investment program which began in 1995, the manufacturing of high tensile steel products: low relaxation steel wire for pre- and post-tensioning and low relaxation steel strand for pre- and post-tensioning.
Fapricela manufactures a range of products that can be considered unusually immense, being proud to be the first company in Portugal to produce low relaxation strand for pre- and post-tensioning.
Throughout its history, the company has always been focusing and combining innovation with high technology (equipment automation), alongside with systematic quality control. Fapricela is at the time a leading wire drawing company in Europe. Production is mainly exported for all over the world.
Fapricela is aware that business strategies should be interwoven with the mission and core values of the organization, and it is becoming increasingly aware that those strategic values lead to long term success when they consist of actions, policies and procedures that are sustainable both for the environment and for the employees. Respect and conservation of our surroundings is therefore a major priority and concern since we are located right next to the estuary of the Mondego River.

Product-related or management system-related certifications:
Fapricela is a certified company on ISO 9001 by APCER/IQNET and CARES UK, with mature procedures of production control and quality compliance. It has also multiple certification schemes related with product certification in countries all over the world such as Spain (AENOR), Portugal (CERTIF), Germany (DIBT –TUM), France (ASQPE), Denmark, Sweeden and Finland (KIWA FI), the Netherlands (KIWA NL), the UK (CARES), Belgium (OCAB/BENOR) Argentina (IRAM-INTI), Poland, Switzerland, Czechia, Austria, etc.

Name and location of production site(s):
Fapricela – Indústria de Trefilaria, S.A.
Apartado 5, Manga da Granja
3060-905 Ançã
Portugal
https://www.fapricela.pt/

Product information

Product name:
Standard Prestressed Steel for Reinforcement of Concrete

Product identification:
Prestressed Steel for Reinforcement of Concrete

Products description:
This EPD represents an average product, weighted according to the production volumes of the following products:

- **Standard PC strand 2/3/7 wires:** low relaxation and smooth surface (crude or oiled) steel strand for all kinds of pre-tensioning with wedge anchoring, such as bars, plates, beams, tubes, pillars and pre-tensioned panels;
- **Standard PC wire:** low relaxation high tensile steel wire for prestressing concrete and all kinds of pre-tensioning with wedge anchoring, such as bars, plates, beams, tubes, pillars and pre-tensioned panels;
- **Standard Bars:** Low relaxation plain or indented wire for posttension of concrete, presented in perfectly straightened bars, cut perpendicularly into precise lengths, to manufacture railway sleepers. Bars can also be supplied with threaded ends.

	Standard PC strand			Standard PC wire/Bars
	2 wires	3 wires	7 wires	
Diameter (mm)	4,5	[4,80 - 8,80]	[6,85 - 18,00]	[3,00 - 10,50]
Section (mm^2)	[7,8 - 8,1]	[11,7 - 39,8]	[27,6 - 203,6]	[6,96 - 88,3]
Linear mass (g/m)	[60,9 - 63,3]	[91,14 - 310,84]	[215,56 - 1590,12]	[54,36 - 689,62]
Curvature (mm/m)	[0 - 25]	[0 - 25]	[0 - 25]	[0 - 25]
Surface	Smooth/Indented	Smooth/Indented	Smooth/Indented	Smooth/Indented
Tensile Strength (Mpa)	>= 1860	[1860 - 2160]	[1770 - 2160]	[1570 - 1860]
Modulus of Elasticity (Gpa)	[185-205]	[185-205]	[185-205]	[195-215]
Maximum Strength (kN)	[14,8 - 17]	[22,1 - 70,1]	[53,9 - 353,2]	[12,6 - 156]
Elastic Limit at 0,1% (kN)	>= 13,0	[19,4 - 47,4]	[47,4 - 246]	[11,1 - 118]
Elastic Limit at 0,2% (kN)	-	[24,6 - 47,5]	[85,1 - 249]	-
Yield Stress 1% Elongation 1 % (kN)	-	[20,5 - 63,1]	[81 - 318]	-
Agt (%)	>= 3,5	>= 2,5	>= 3,5	>= 3,5

UN CPC code:
421 Structural metal products and parts thereof.

Geographical scope:
The LCA study was carried out according to the European scope.
Product Market: Global

LCA information

Declared unit:
1 kg of standard prestressed steel for reinforcement of concrete (packaging included).

Reference service life:
The service life of the building (50 years) was considered, since, once installed, the product is protected by other elements and does not require maintenance.

Time representativeness:
2022

Database(s) and LCA software used:
- Ecoinvent v3.9.1
- SimaPro v9.5

Data Quality:
Primary data was provided by Fapricela and was based on the average production of standard prestressed steel for reinforcement of concrete in 2022. For processes which the producer had no influence on or no specific information about, such as extraction of raw materials, production of customised products and electricity generation, literature and generic data from ecoinvent database v3.9.1 was used, considering geographical and temporal significance.

Cut-off rules:
Criteria for the exclusion of inputs and outputs follow the requirements of EN 15804: 2012+A2 2019 and ISO 21930:2017. Where there was insufficient data for a unit process in the LCA study, the cut-off criteria were set at 1% of renewable and non-renewable primary energy usage and 1% of the total mass input of that process, with a maximum of 5% of energy usage and mass per module, ensuring that at least 95% of the environmental impact per module is considered. Based on this cut-off criteria, ancillary materials (such as expendable components and chemicals for wastewater treatment, etc.) were excluded from the analysis.

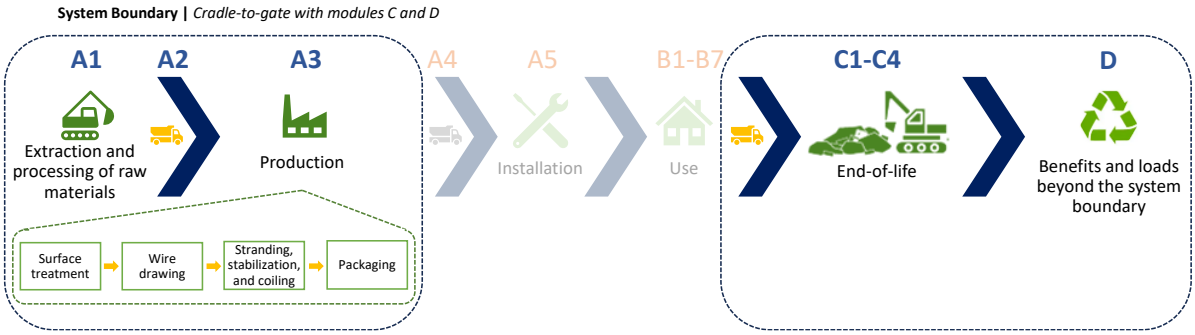
Allocation:
During the production of special prestressed steel products, no co-products are produced. Therefore, allocation does not apply. Furthermore, all products sent to recycling are treated under the “polluter pays” principle.

Electricity Sources:

Electricity used in module A3 was modelled using the Portuguese energy grid residual mix from ecoinvent v3.9.1.

Type of information	Description
Electricity mix	Hard coal: 2%; Hydro: 16%; Natural gas: 25%; Oil: 3%; Wind: 17%; Biomass: 4%; PV: 4%; European attribute mix: 29%
GWP-GHG (kg CO_2 eq/kWh)	0,343

Description of system boundaries:
Cradle-to-gate with modules C and D. This EPD covers the information module A1-A3, C1-C4 and D, comprising of the following modules: [A1] raw material extraction and processing, processing of secondary material input; [A2] transport to the manufacturer; [A3] manufacturing; [C2] transport to waste processing; [C3] waste processing for reuse, recovery and/or recycling; [C4] disposal and module D.



Product stage [A1-A3]:
Modules A1-A3 cover the extraction, production and acquisition of the main raw materials and pre-products, as well as electricity generation, transport of all raw materials considered in module A1 to the factory gate, and production of the final products, including waste management.

Construction process stage [A4-A5]:
The construction process stage is not included.

Use stage [B1-B7]:
The use stage is not included.

End of life stage [C1-C4]:
A scenario of 90% recycling and 10% landfilling of waste is considered, according to prEN 17662:2021.

Module C1:
This module includes the demolition of the product from the building, including initial on-site sorting of materials. It was assumed that all products are used as concrete reinforcement.

Module C2:
This module includes the transportation of the waste to a waste processing facility, recycling facility and final disposal, according to the parameters detailed in the table below.

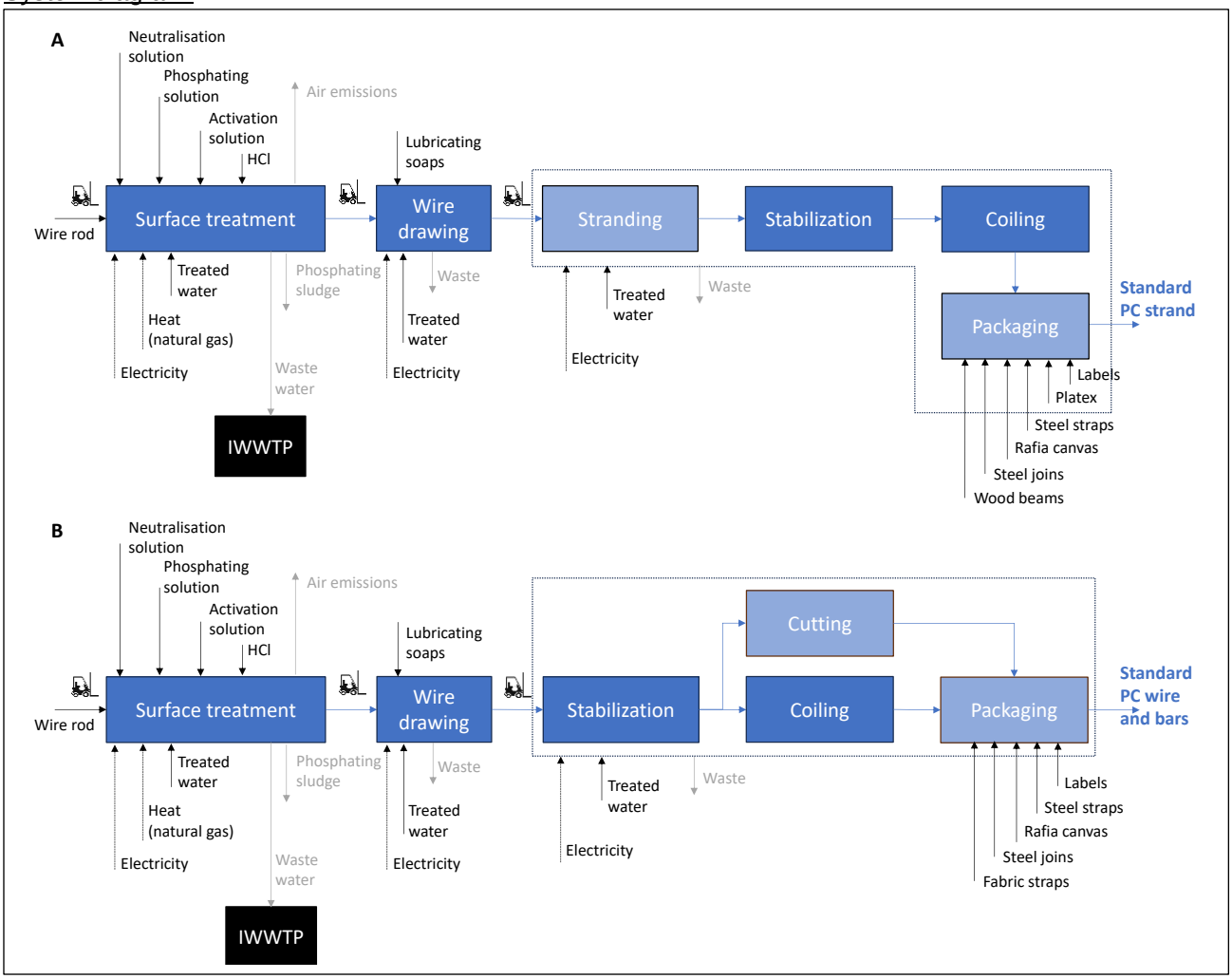
Module C3:
This model includes waste processing for recycling.

Module C4:
Module C4 includes the landfill disposal of the waste.

Resource recovery stage [D]:
This module includes the net benefits and loads from net flows of secondary materials leaving the product system. In the recycling process, an efficiency of 95% was taken into account.

Module	Parameters	Standard prestressed steel
De-construction, demolition [C1]	Diesel consumption (L/t)	13,83 (prEN 17662)
Transport to recovery/disposal [C2]	Distance to scrap processing plant (km)	100 (prEN 17662)
	Distance to end of life (km)	200 (prEN 17662)
	Distance to recycling plant (km)	100 (prEN 17662)
	Type of transport	Truck
Waste processing for reuse, recovery and/or recycling [C3]	End-of-life scenario	90% recycling (prEN 17662)
	Electricity consumption for sorting and shredding (kWh/t)	3,7
Final disposal [C4]	End-of-life scenario	10% landfill (prEN 17662)
Benefits and loads beyond the system boundary [D]	Steel recycling rate	90% (prEN 17662)
	Steel recycling efficiency	95% (prEN 17662)

System diagram:



Production process of: A) standard PC strand; and B) standard PC wire and bars.

The industrial process consists of four main processes, which are common to all the standard prestressed steel products: surface treatment; wire drawing, stabilization, and packaging. For the PC strand, before stabilization, there is also stranding. For Bars, PC wire is cut before being packed (there is no coiling).

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results):

	Product stage			Construction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	ND	ND	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	EU	EU	PT	-	-	-	-	-	-	-	-	-	EU				EU
Specific data used	>60%[a]			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	<10%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-

x: included / ND: not declared / PT: Portugal / EU: European.

[a] The percentage of specific data is assumed to be larger than 60%, but it cannot be proved since one EPD used as data source lack information on the percentage of specific data used.

Content information

Standard prestressed steel (average product)*

Product components	Weight, kg	Post-consumer material, weight-%	Biogenic material, weight (C)		Biogenic material, weight (CO_2)	
			%	kg C/kg	%	kg CO_2/kg
Wire rod (steel)	1.00E00	54	0	0	0	0
Lubricating oil	7,27E-06	0	0	0	0	0
TOTAL	1.00E00	54	0	0	0	0

Packaging materials	Weight, kg	Weight-% (versus the product)	Weight biogenic carbon, kg C/kg	Weight biogenic carbon, kg CO_2/kg
Raffia canvas	1,85E-04	0,02	9,82E-05	3,60E-04
Steel straps	1,23E-03	0,12	0	0
Steel joins	9,75E-04	0,10	0	0
Labels	6,49E-05	0,01	0	0
Wood beams	2,21E-03	0,22	1,09E-03	4,36E-03
Fabric straps	2,80E-06	0,00	0	0
TOTAL	4,67E-03	0,47	1,18E-03	4,36E-03

*The product does not contain any substance included in the Candidate List of Substances of Very High Concern (SVHCs) for authorization with concentrations higher than 0.1% weight by weight (w/w).

Results of the environmental performance indicators

The life cycle impact assessment (LCIA) results represent an average product and were calculated based on a weighted average taking into account the production of each type of products (standard PC strand, PC wire and Bars) in 2022. LCIA results are relative expressions and do not predict impacts on category endpoints, the exceeding of thresholds, safety margins or risks. The use of the results of modules A1-A3 without considering the results of module C is discouraged. A1-A3 results include the "balancing-out reporting" of biogenic CO_2 of packaging released in module A5.

Mandatory impact category indicators according to EN 15804 (based on EF 3.1)

Results per declared unit (1 kg)							
Indicator	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-fossil	kg CO_2 eq.	9,31E-01	4,70E-02	3,21E-02	1,27E-03	2,70E-04	-3,37E-01
GWP-biogenic	kg CO_2 eq.	1,21E-03	8,74E-06	9,69E-06	3,48E-06	1,06E-07	7,68E-03
GWP-luluc	kg CO_2 eq.	1,72E-03	1,92E-06	6,28E-07	8,89E-06	1,36E-08	-5,61E-05
GWP-total	kg CO_2 eq.	9,34E-01	4,70E-02	3,21E-02	1,28E-03	2,70E-04	-3,30E-01
ODP	kg CFC 11 eq.	2,59E-08	7,41E-10	6,93E-10	3,51E-11	4,02E-12	-8,28E-09
AP	mol H^+ eq.	3,01E-03	4,50E-04	8,35E-05	5,19E-06	2,51E-06	-1,23E-03
EP-freshwater	kg P eq.	2,16E-04	3,54E-07	2,19E-07	2,27E-07	8,07E-09	-1,39E-04
EP-marine	kg N eq.	7,25E-04	2,11E-04	3,27E-05	8,44E-07	1,14E-06	-2,85E-04
EP-terrestrial	mol N eq.	7,22E-03	2,30E-03	3,46E-04	9,35E-06	1,24E-05	-3,04E-03
POCP	kg NMVOC eq.	3,75E-03	6,76E-04	1,35E-04	3,38E-06	3,71E-06	-1,78E-03
ADP-minerals&metals*	kg Sb eq.	4,52E-07	1,97E-09	1,10E-09	2,27E-11	1,07E-11	-5,64E-08
ADP-fossil*	MJ	1,16E+01	6,18E-01	4,26E-01	2,09E-02	3,47E-03	-3,55E+00
WDP*	m^3	3,08E-01	7,92E-04	3,90E-04	4,42E-04	4,78E-06	-1,60E-02
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption						

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

Additional mandatory and voluntary impact category indicators

Results per declared unit (1 kg)							
Indicator	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG[1]	kg CO_2 eq.	9,32E-01	4,70E-02	3,21E-02	1,28E-03	2,70E-04	-3,37E-01
Additional voluntary indicators e.g. the voluntary indicators from EN 15804 or the global indicators according to ISO 21930:2017							

Resource use indicators

Results per declared unit (1 kg)							
Indicator	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	1,59E+00	1,20E-03	1,12E-03	7,57E-03	1,55E-05	-3,86E-02
PERM	MJ	3,53E-02	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
PERT	MJ	1,63E+00	1,20E-03	1,12E-03	7,57E-03	1,55E-05	-3,86E-02
PENRE	MJ	1,54E+01	6,18E-01	4,26E-01	7,26E-03	3,47E-03	-3,55E+00
PENRM	MJ	2,56E-02	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
PENRT	MJ	1,55E+01	6,18E-01	4,26E-01	2,09E-02	3,47E-03	-3,55E+00
SM	kg	8,83E-01	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
RSF	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
NRSF	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
FW	m^3	1,23E-02	3,07E-05	1,78E-05	1,36E-05	1,82E-07	-5,07E-04
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water						

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.

Waste indicators

Results per declared unit (1 kg)							
Indicator	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed	kg	6,61E-05	4,13E-06	2,81E-06	5,86E-08	2,23E-08	-4,05E-05
Non-hazardous waste disposed	kg	1,44E-01	4,59E-05	2,10E-05	1,41E-05	9,99E-02	-4,76E-04
Radioactive waste disposed	kg	2,11E-05	3,01E-08	3,66E-08	5,84E-08	2,06E-10	-8,65E-07

Output flow indicators

Results per declared unit (1 kg)							
Indicator	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Components for re-use	kg	1,29E-01	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Material for recycling	kg	3,19E-02	0,00E+00	0,00E+00	9,00E-01	0,00E+00	0,00E+00
Materials for energy recovery	kg	5,13E-04	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Exported energy, electricity	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Exported energy, thermal	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00

References

- International Organization for Standardization (ISO). EN ISO 14040:2006. Environmental management – life cycle assessment –principles and framework, 2006.
- International Organization for Standardization (ISO). EN ISO 14044:2006. Environmental management – Life cycle assessment – Requirements and guidelines, 2006.
- International Organization for Standardization (ISO). ISO 14025:2006. Environmental labels and declarations — Type III environmental declarations — Principles and procedures, 2006.
- European Standard. EN 15804:2012+A2:2019. Sustainability of construction works - Environmental product declarations - Core rules for the product category of construction products, 2019.
- European Standard. CEN/TR 15941:2010. Sustainability of construction works - Environmental product declarations - Methodology for selection and use of generic data, 2010.
- International Organization for Standardization (ISO). ISO 21930:2017 – Sustainability in buildings and civil engineering works — Core rules for environmental product declarations of construction products and services, 2017.
- General Programme Instructions of the International EPD® System. Version 4.0.
- PCR 2019:14 Construction Products, Version 1.3.3.
- prEN 17662:2021 Execution of steel structures and aluminium structures - Environmental Product Declarations - Product category rules complementary to EN 15804 for Steel, Iron and Aluminium structural products for use in construction works.
- Celsa Group. Environmental Product Declaration of Hot rolled wire rod and reinforcing steel products. The International EPD System. S-P-07496.

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